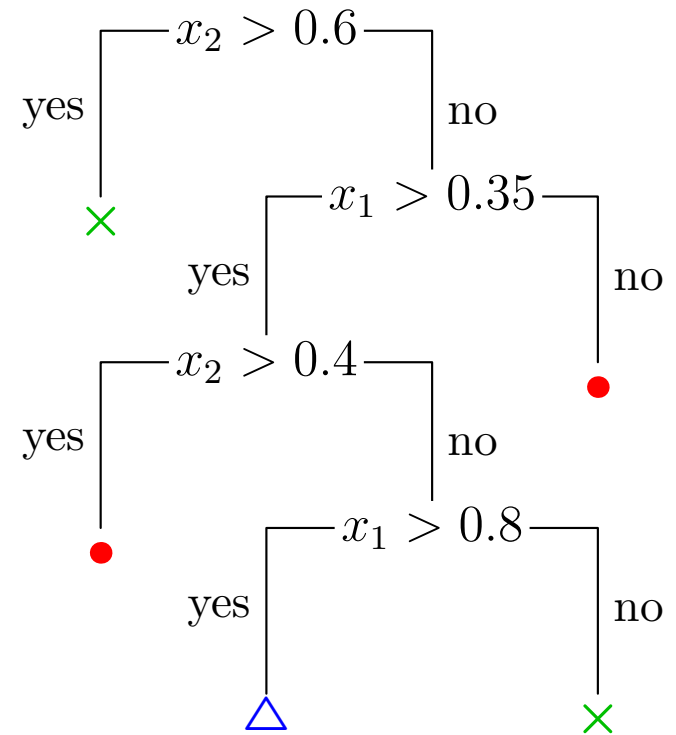
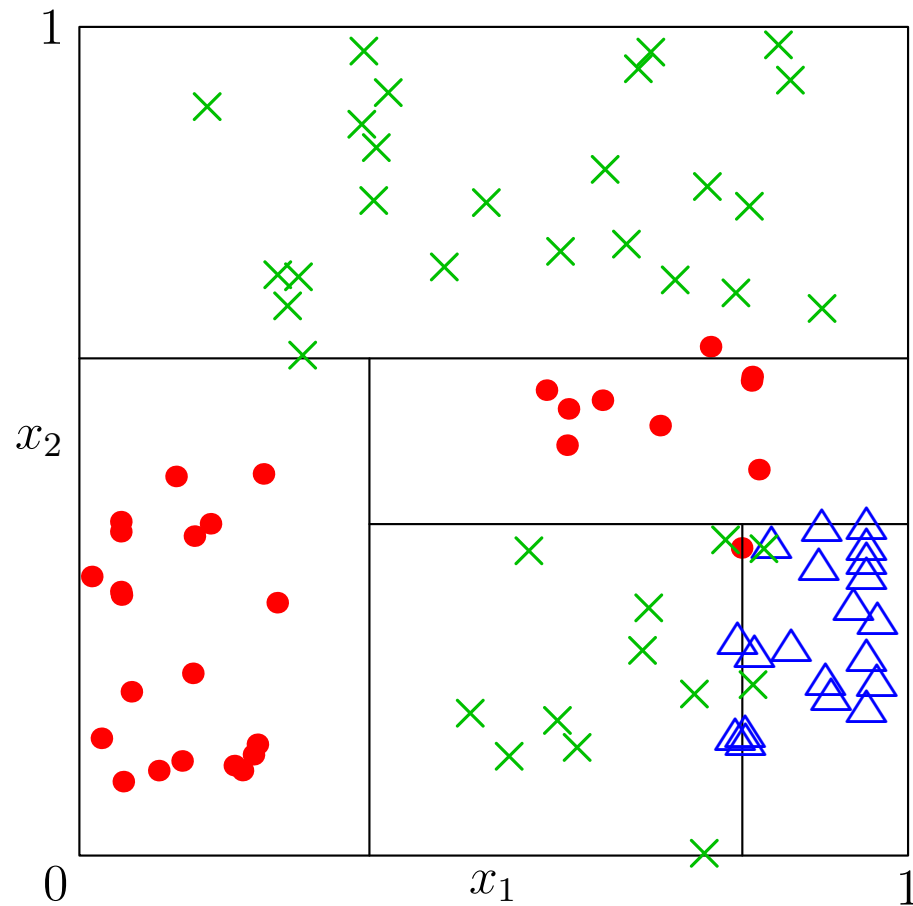


Advanced Machine Learning

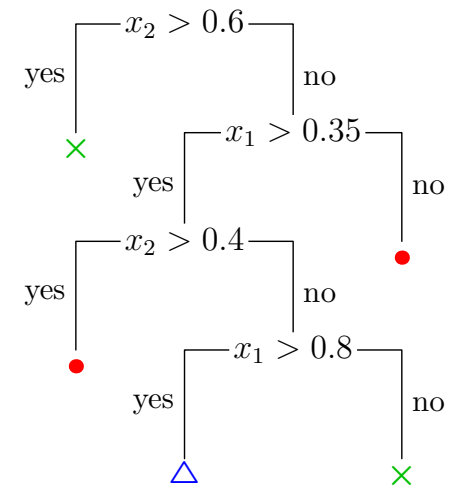
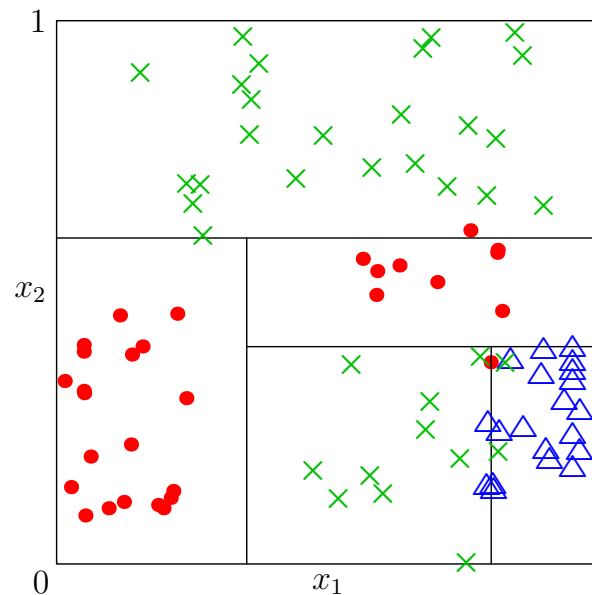
Boosting



Boosting, AdaBoost, Gradient Boosting

Outline

1. **Boosting**
2. AdaBoost
3. Gradient Boosting
4. Dropout



Boosting

- In boosting we make a **strong learner** by using a weighted sum of **weak learners**

$$C_n(\mathbf{x}) = \sum_{i=1}^n \alpha_i \hat{h}_i(\mathbf{x})$$

- Weak learners, $\hat{h}_i(\mathbf{x})$, are learning machine that do a little better than chance
- The trick is to choose the weights, α_i
- Because the weak learners do little better than chance we (miraculously) **don't** overfit

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Shallow Trees

- One of the most effective type of weak learner are very shallow trees
- Sometimes we just use one variable (the stump)
- There are different algorithms for choosing the weights
 - ★ adaboost
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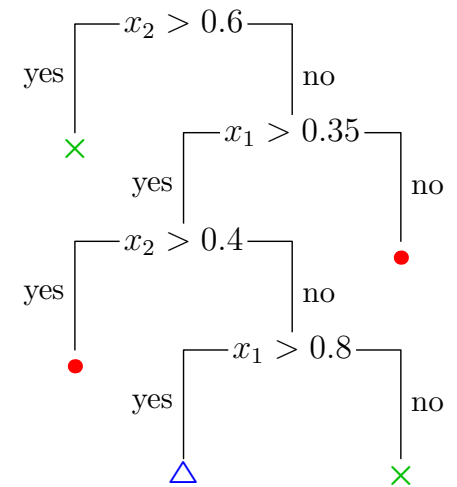
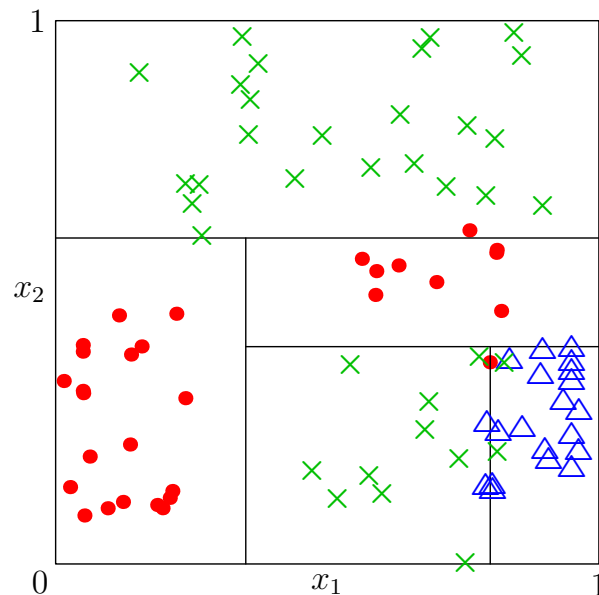
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 - ★ adaboost—a classic algorithm for binary classification
 - ★ gradient boosting—used for regression, trains a weak learner on the residual errors

Outline

1. Boosting
2. **AdaBoost**
3. Gradient Boosting
4. Dropout



Boosting a Binary Classifier

- Suppose we have a binary classification task with data $\mathcal{D} = \{(\mathbf{x}^\mu, y^\mu) | \mu = 1, 2, \dots, m\}$ with $y^\mu \in \{-1, 1\}$
- Our i^{th} weak learner provides a prediction $\hat{h}_i(\mathbf{x}^\mu) \in \{-1, 1\}$
- We ask, can we find a linear combination

$$C_n(\mathbf{x}) = \alpha_1 \hat{h}_1(\mathbf{x}) + \alpha_2 \hat{h}_2(\mathbf{x}) + \dots + \alpha_n \hat{h}_n(\mathbf{x})$$

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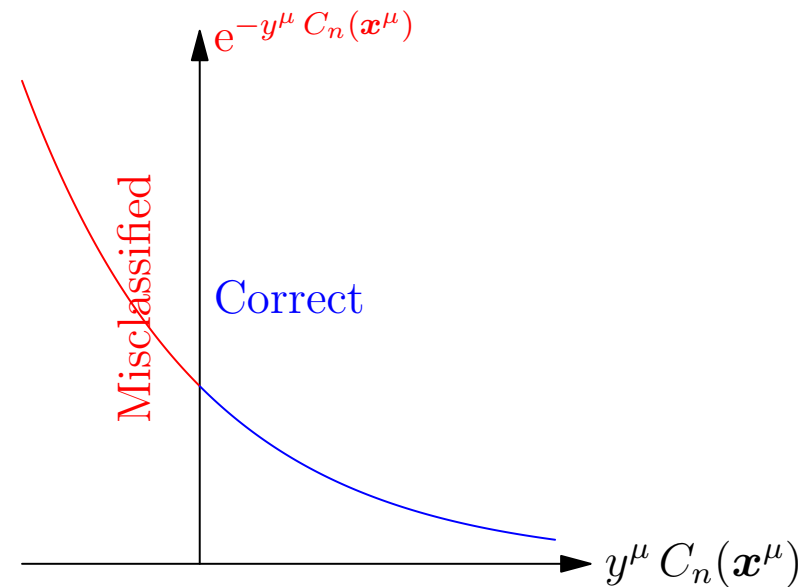
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- Note we want $y^\mu C_n(\mathbf{x}^\mu) > 0$

AdaBoost

- AdaBoost is a classic solution to this problem
- It assigns an “loss function”

$$L_n = \sum_{\mu=1}^m e^{-y^\mu C_n(\mathbf{x}^\mu)}$$

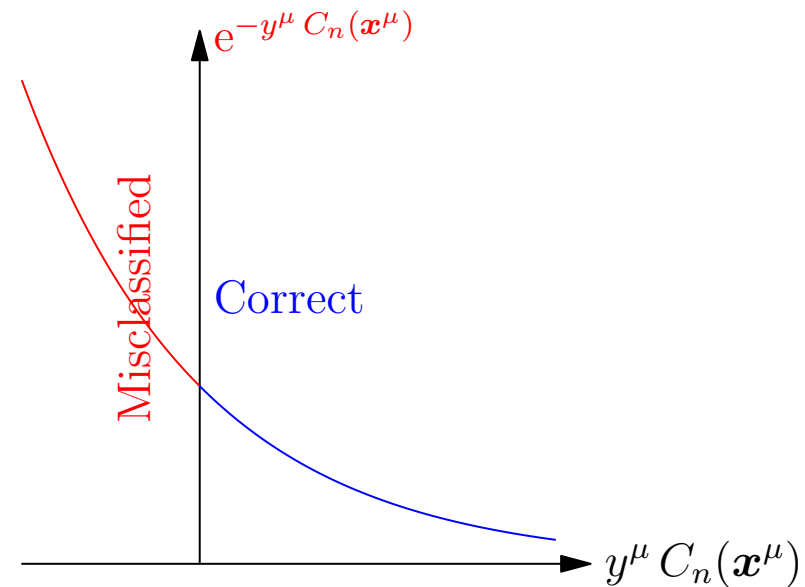


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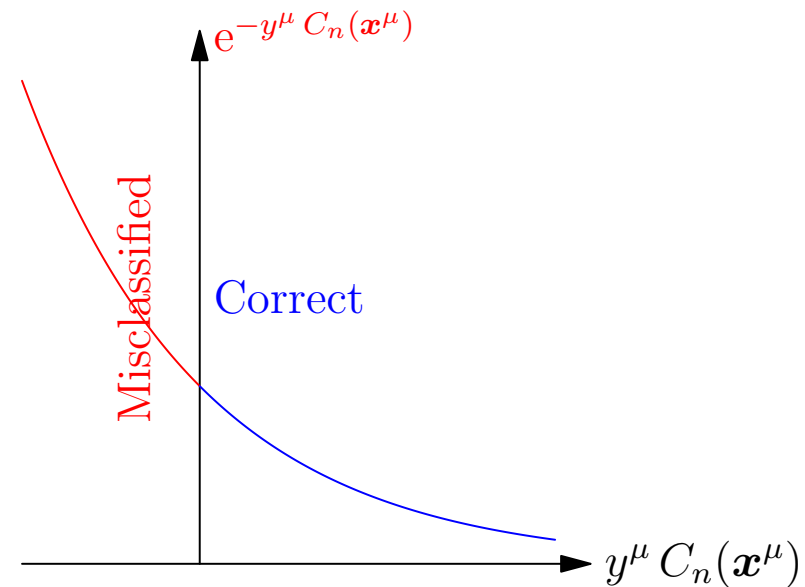


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- We build up a strong learner iteratively (greedily)

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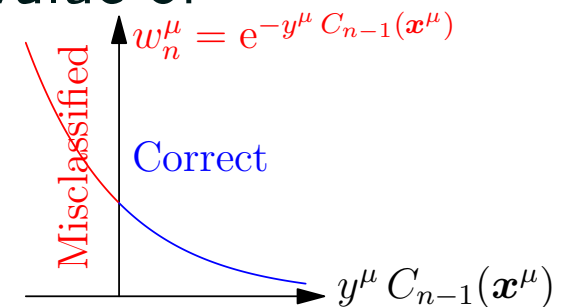
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- To minimise the loss

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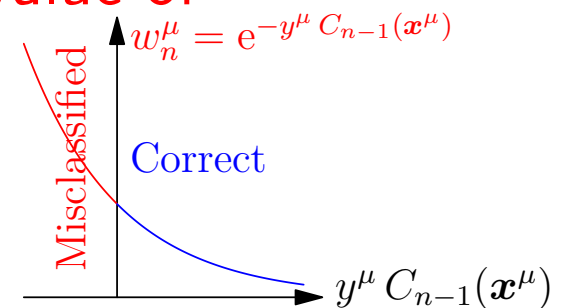
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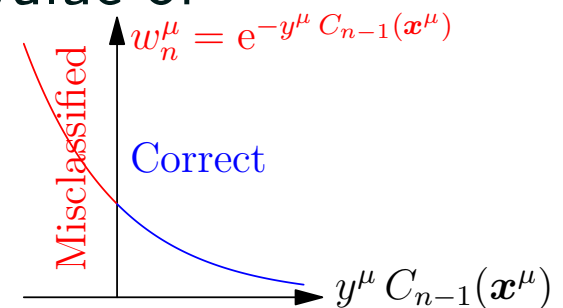
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Algorithm

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2. Associate a weight, w_n^μ , with every data point $(\mathbf{x}^\mu, y^\mu)$, $\mu = 1, 2, \dots, m$
3. Initially $w_1^\mu = 1$
4. Choose the weak learning, $\hat{h}_n(\mathbf{x}) \in \mathcal{W}$, that minimises $\sum_{\mu: y^\mu \neq \hat{h}_n(\mathbf{x}^\mu)} w_n^\mu$
5. Update predictor $C_n(\mathbf{x}) = C_{n-1}(\mathbf{x}) + \alpha_n \hat{h}_n(\mathbf{x})$ where
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2. Associate a weight, w_n^μ , with every data point $(\mathbf{x}^\mu, y^\mu)$, $\mu = 1, 2, \dots, m$
3. Initially $w_1^\mu = 1$ (large weight, w_n^μ , means $(\mathbf{x}^\mu, y^\mu)$ is poorly classified)
4. Choose the weak learning, $\hat{h}_n(\mathbf{x}) \in \mathcal{W}$, that minimises
$$\sum_{\mu: y^\mu \neq \hat{h}_n(\mathbf{x}^\mu)} w_n^\mu$$
5. Update predictor $C_n(\mathbf{x}) = C_{n-1}(\mathbf{x}) + \alpha_n \hat{h}_n(\mathbf{x})$ where
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Performance

- Adaboost works well with weak learners, usually out-performing bagging
- It doesn't work well with strong learners (tends to over-fit)
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- In contrast **gradient boosting** used for regression is very popular

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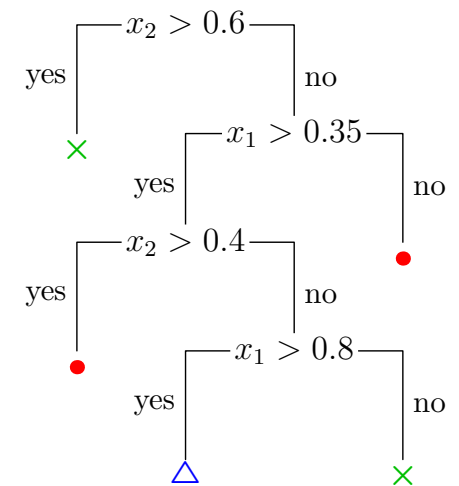
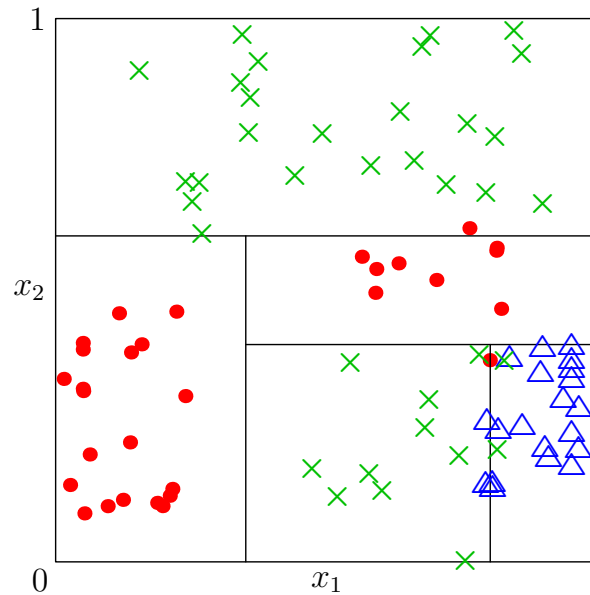
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2. AdaBoost
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Gradient Boosting

- In gradient boosting we again build a strong learner as a linear combination of weak learners

$$C_n(\mathbf{x}) = C_{n-1}(\mathbf{x}) + \hat{h}_n(\mathbf{x})$$

- Gradient boosting used on regression (again using decision trees)
- At each step $\hat{h}_n(\mathbf{x})$ is trained to predict the **residual error**, $\Delta_{n-1} = y - C_{n-1}(\mathbf{x})$, (i.e. the target minus the current prediction)
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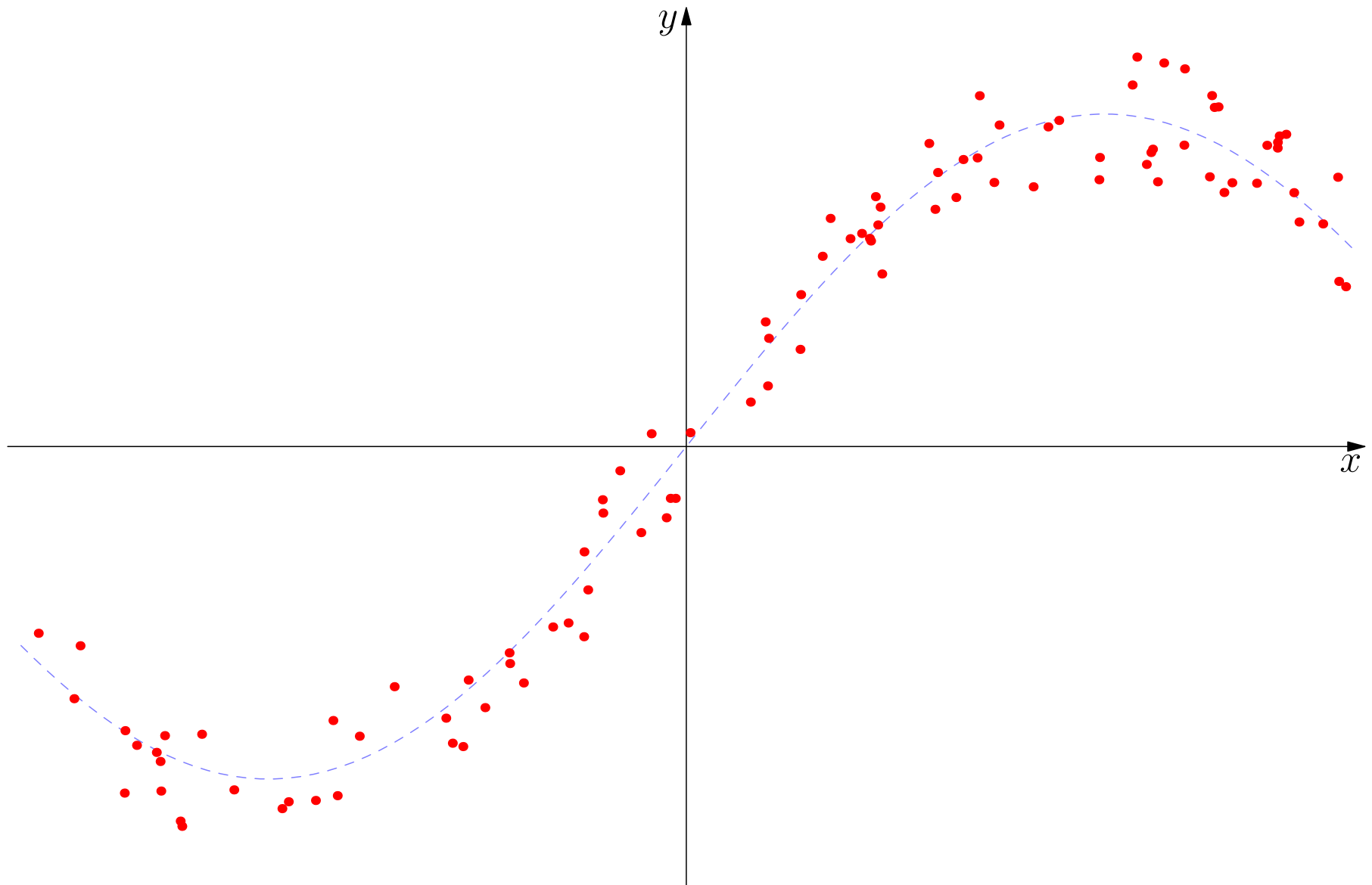
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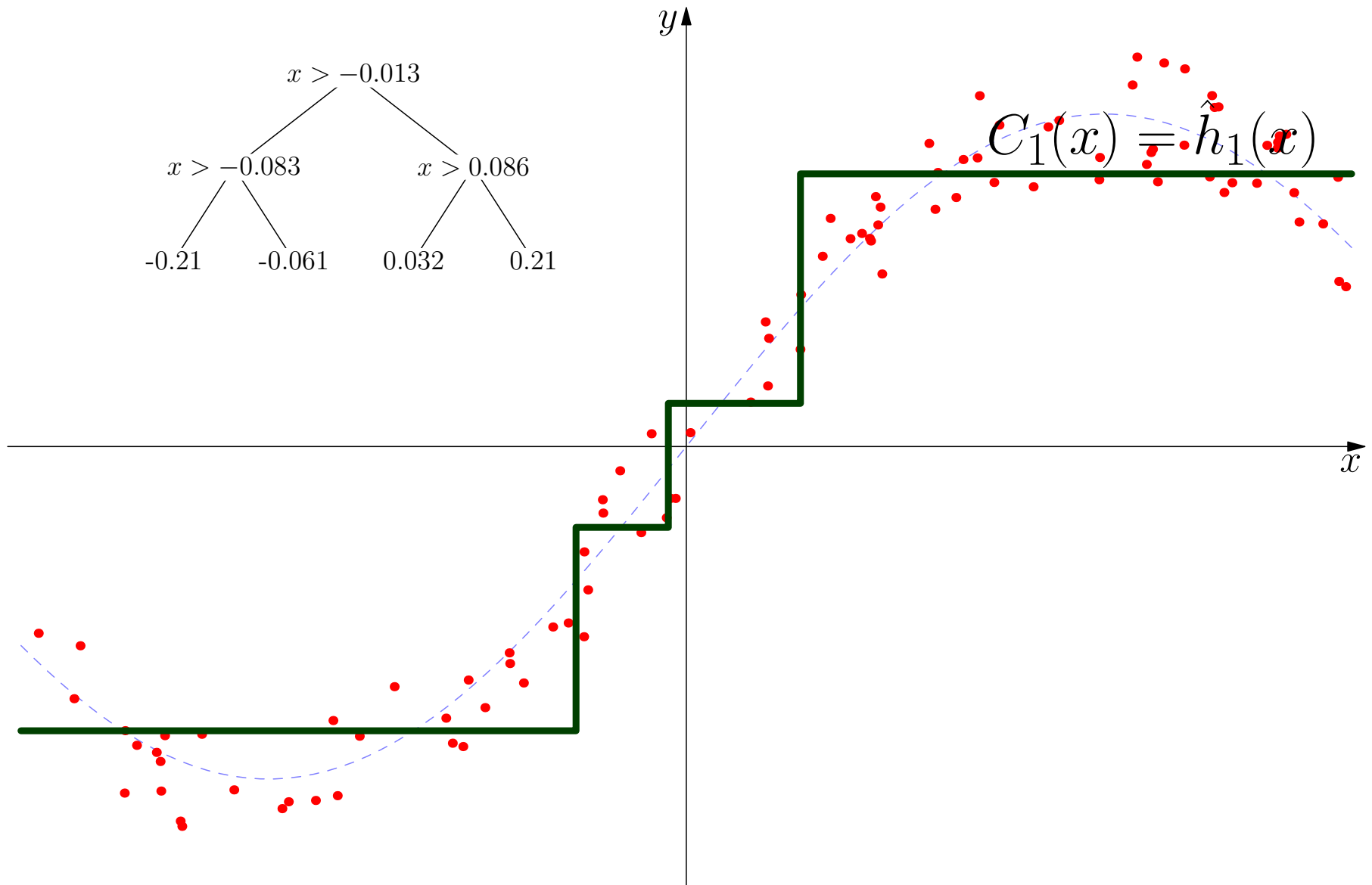
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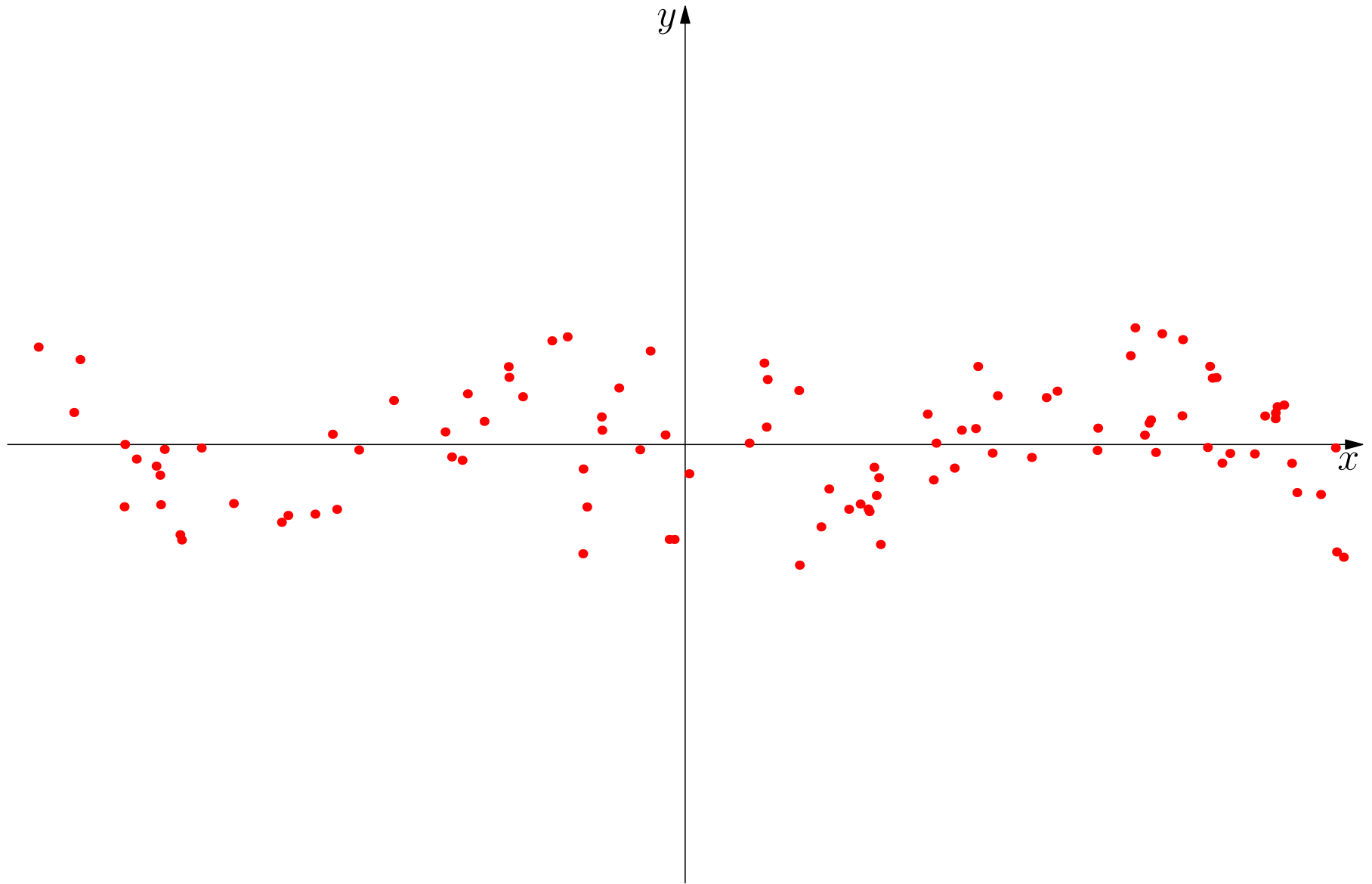
Fitting a Sin Wave



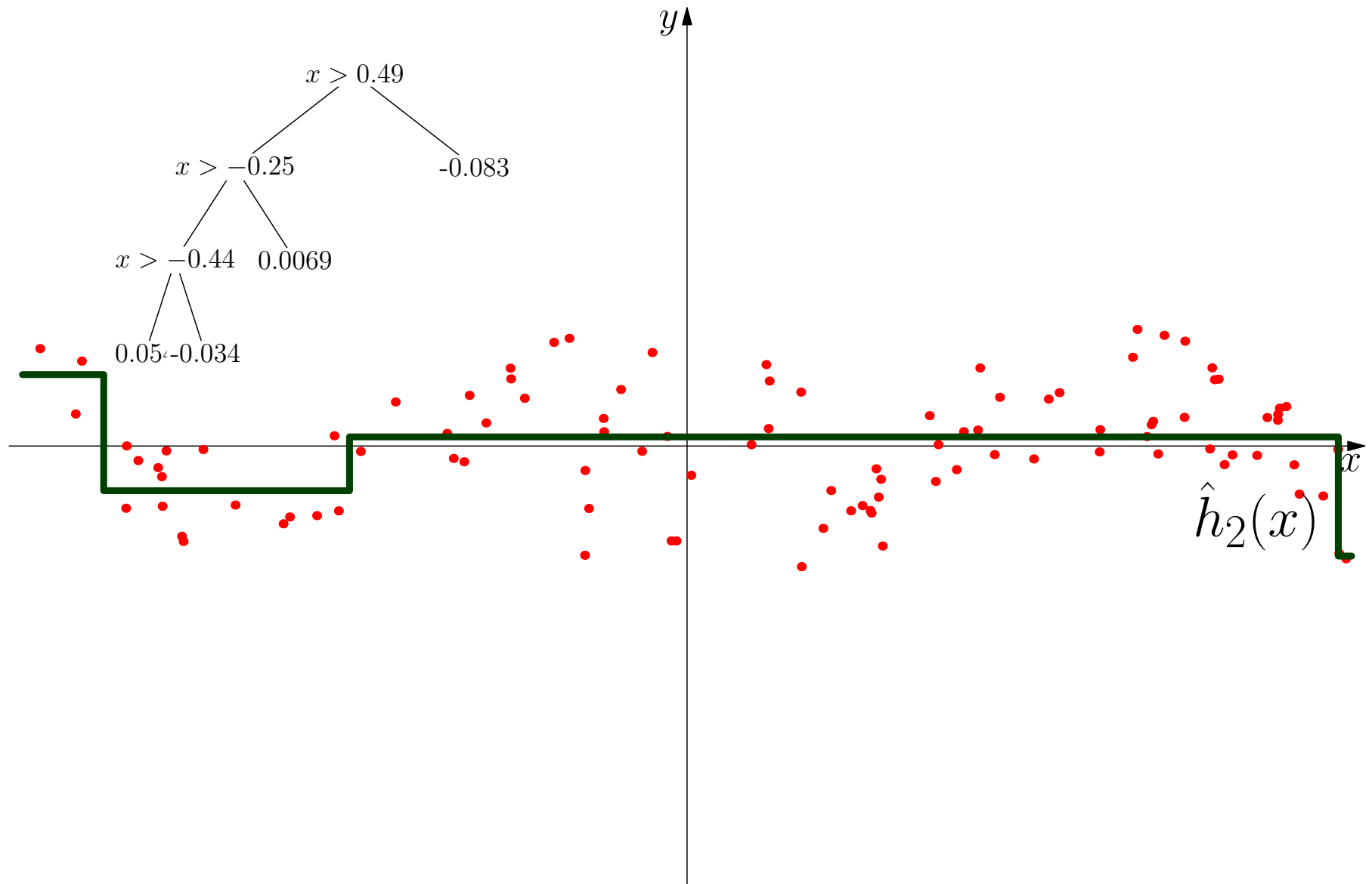
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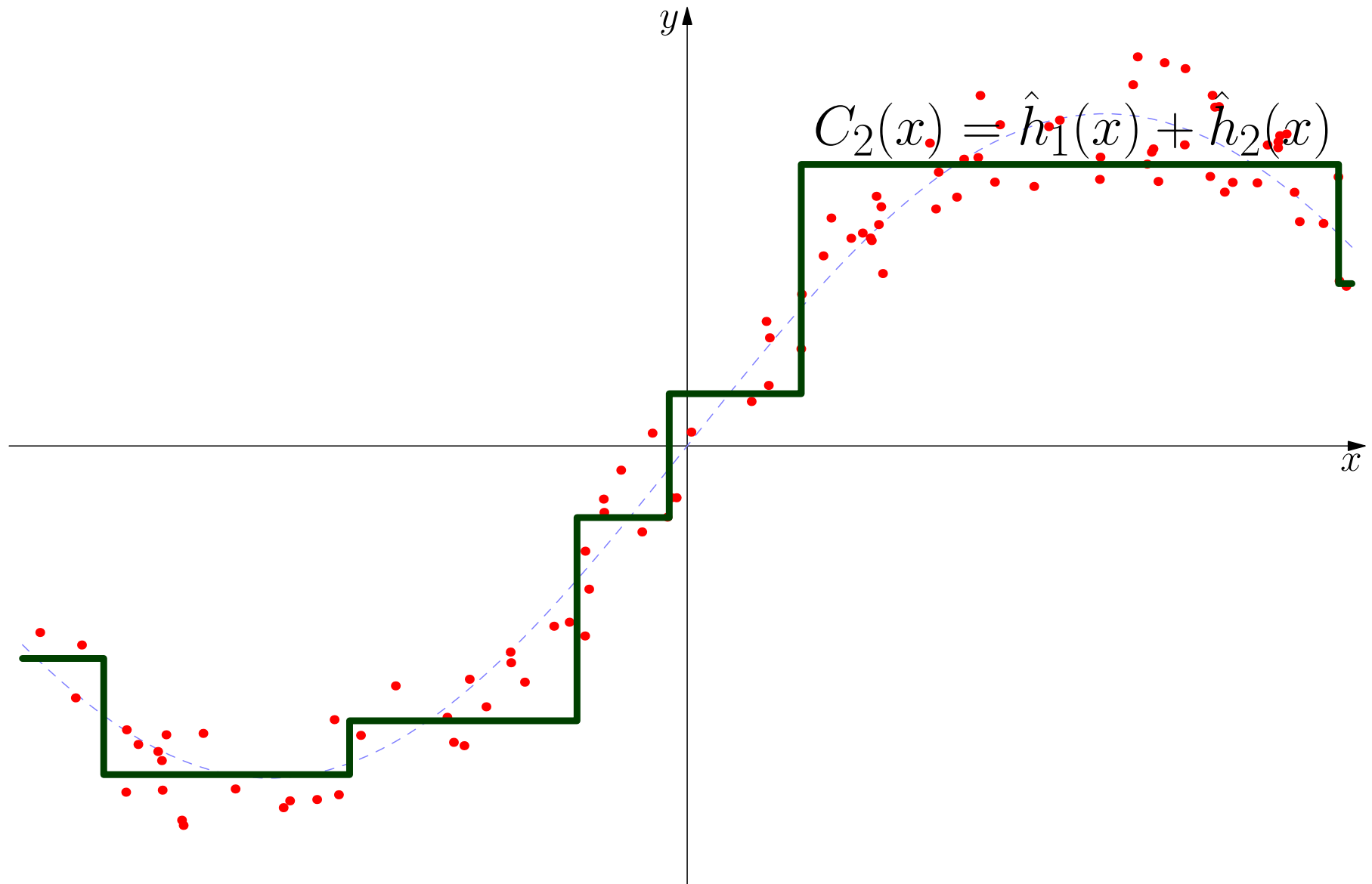
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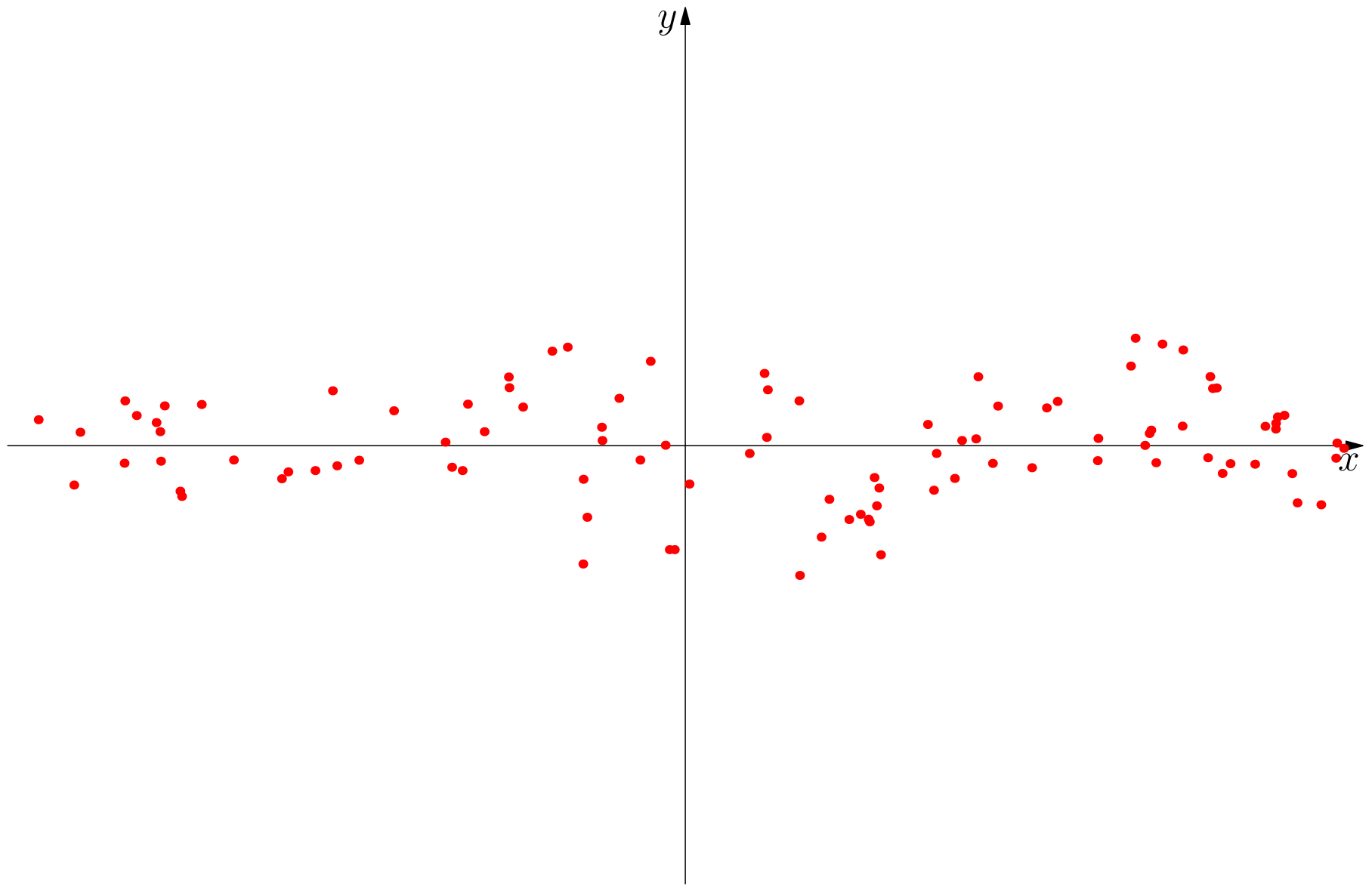
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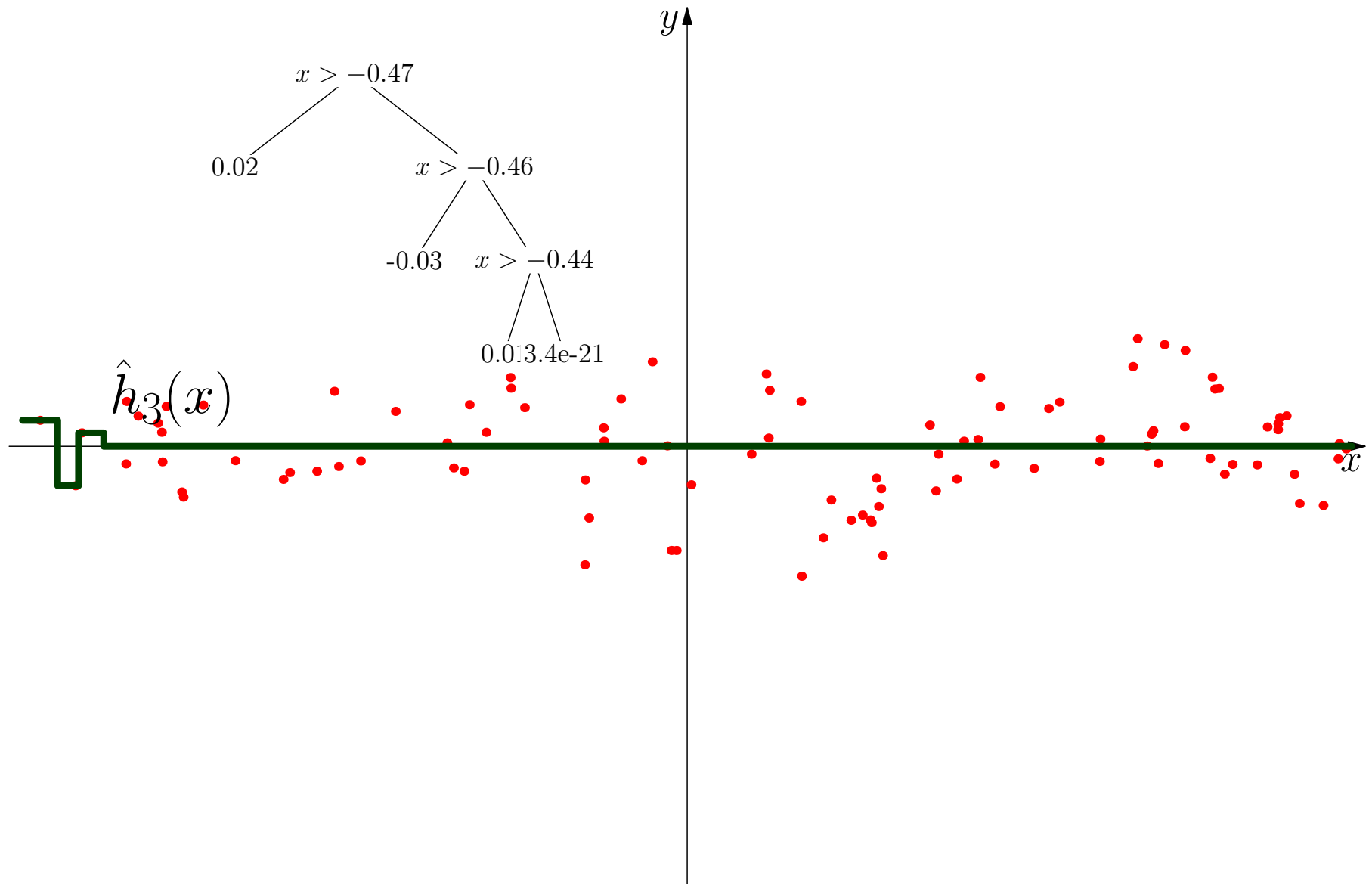
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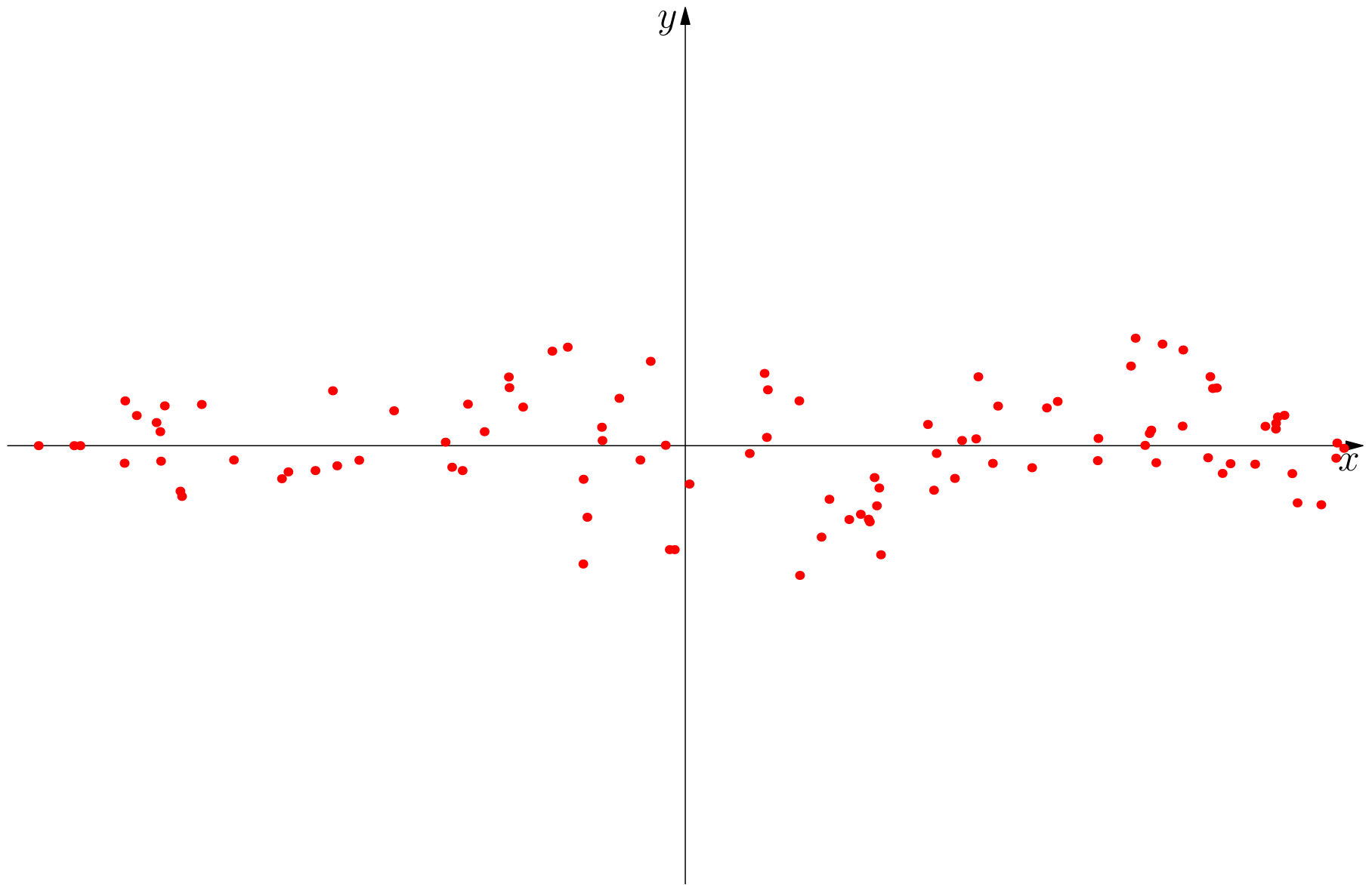
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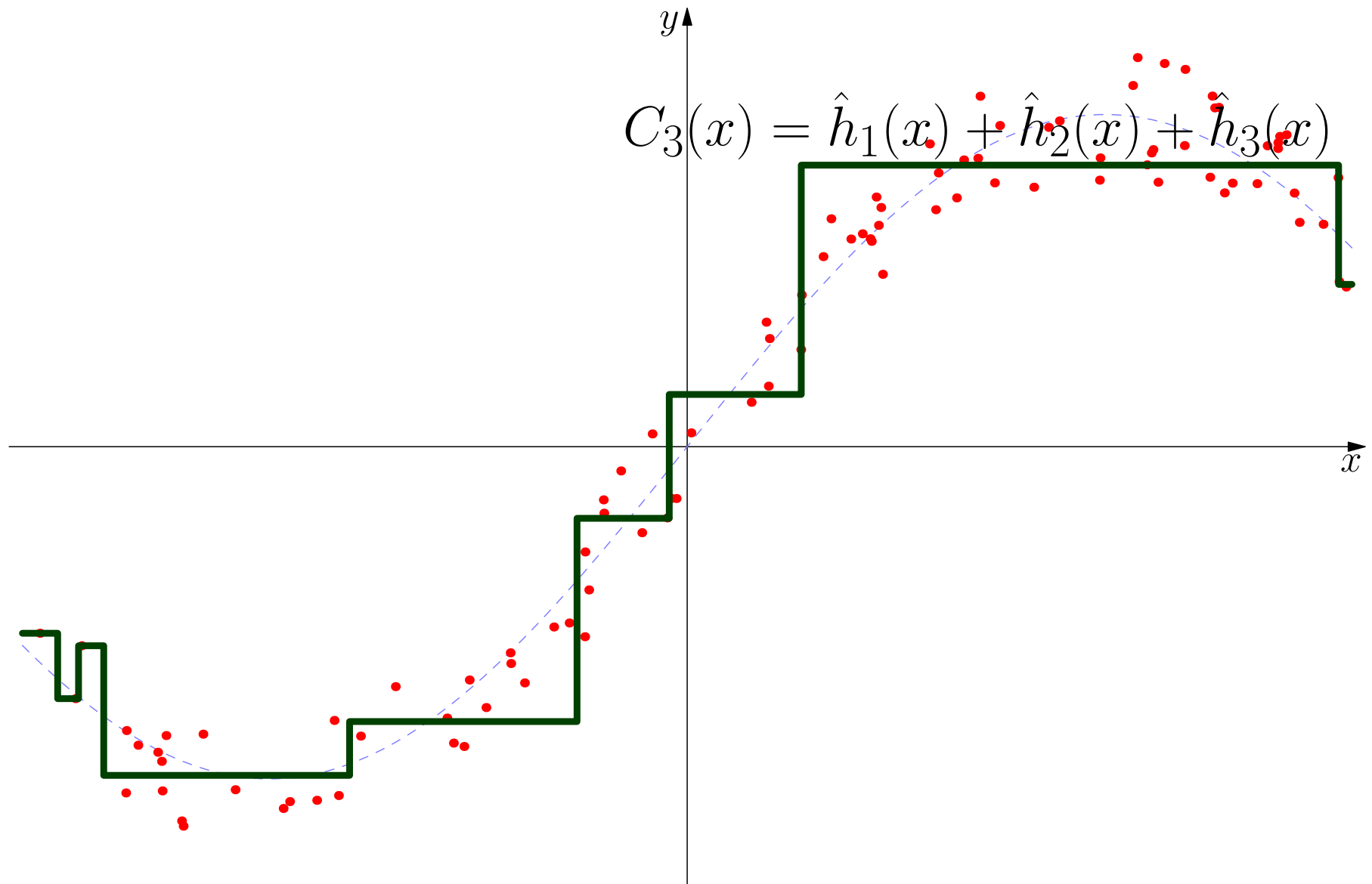
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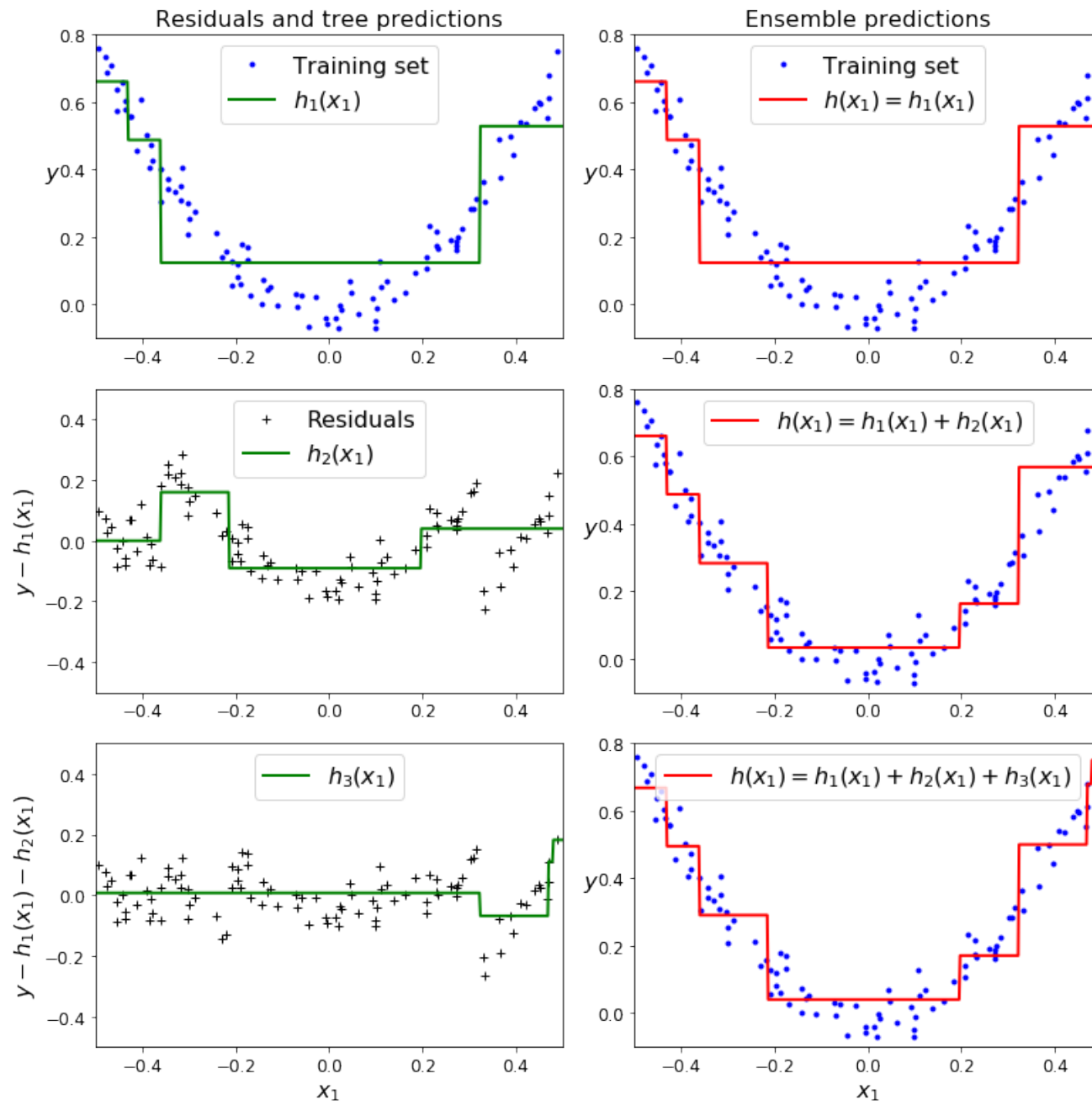


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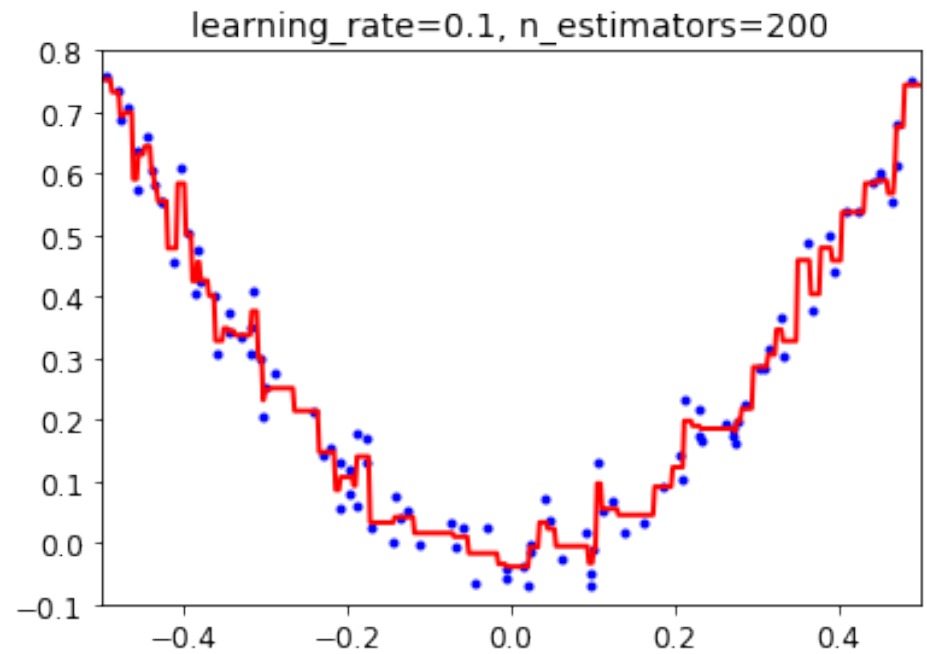
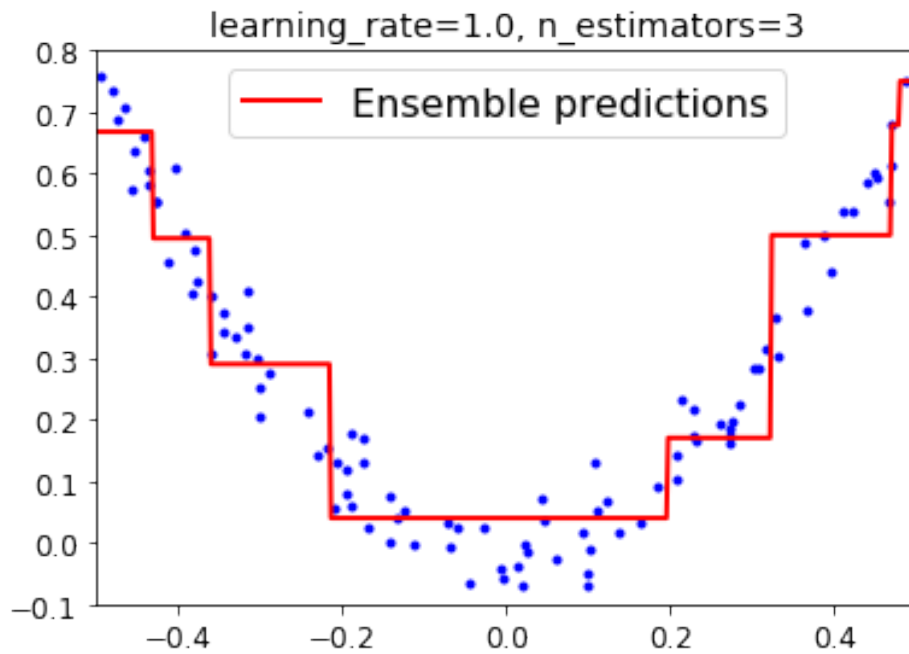
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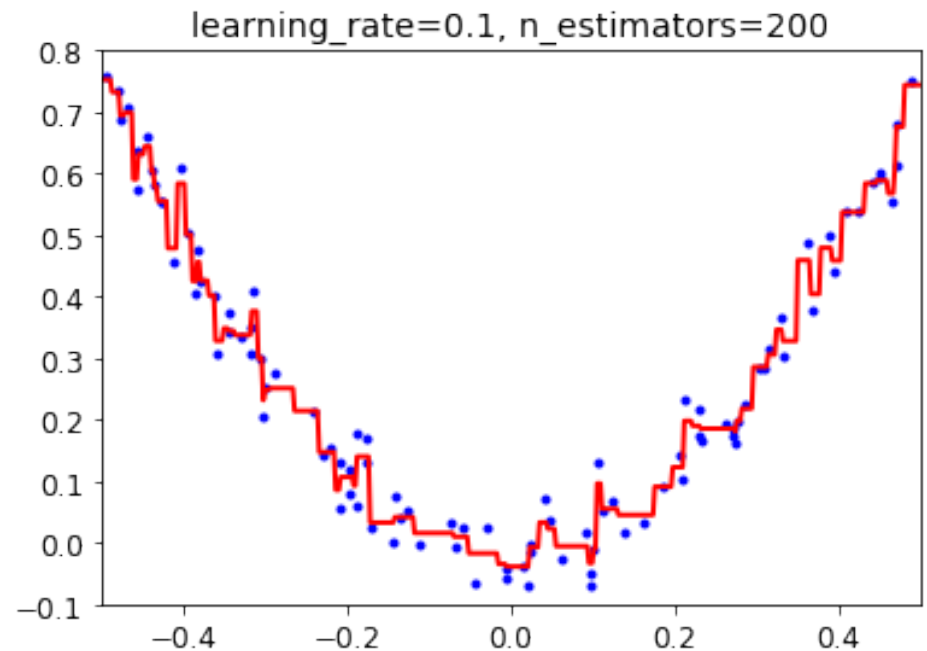
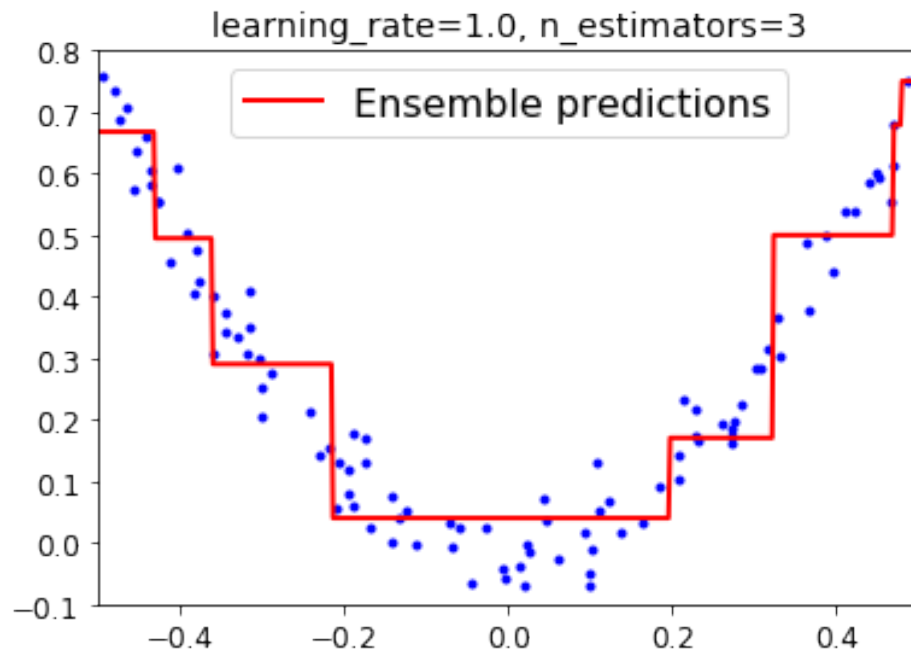
Keep On Going

- We can keep on going



Keep On Going

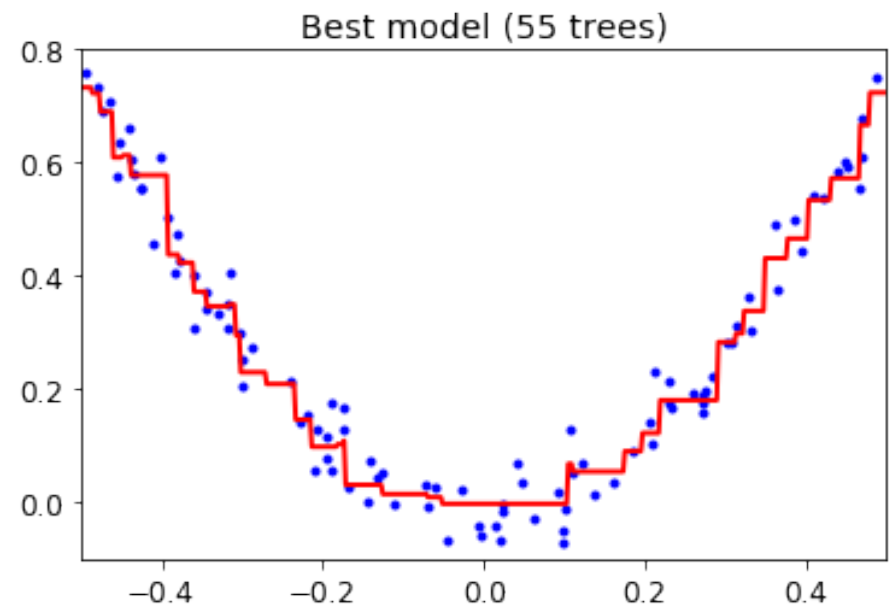
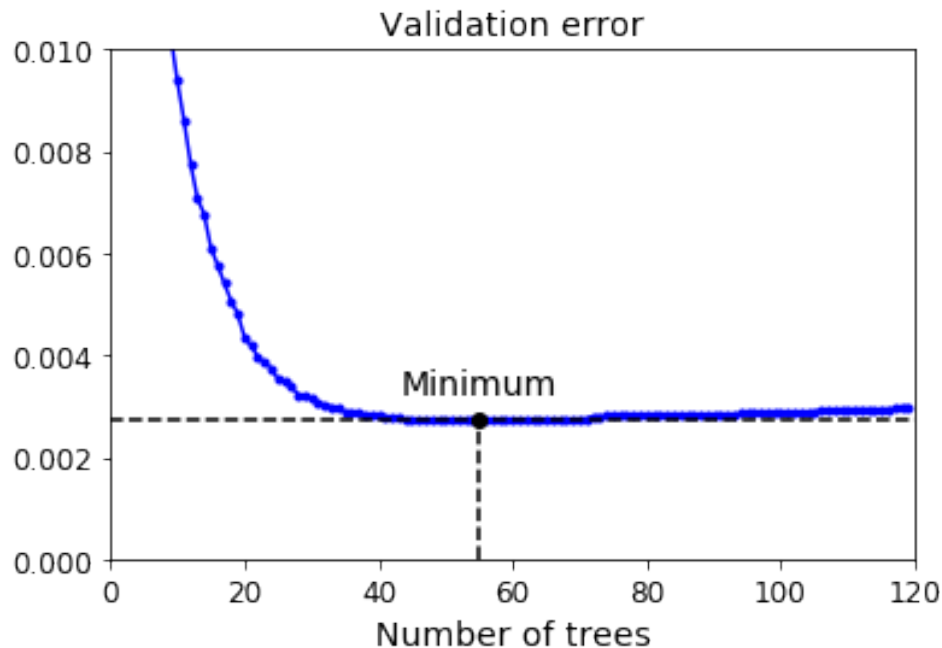
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- But we will over-fit eventually

Early Stopping

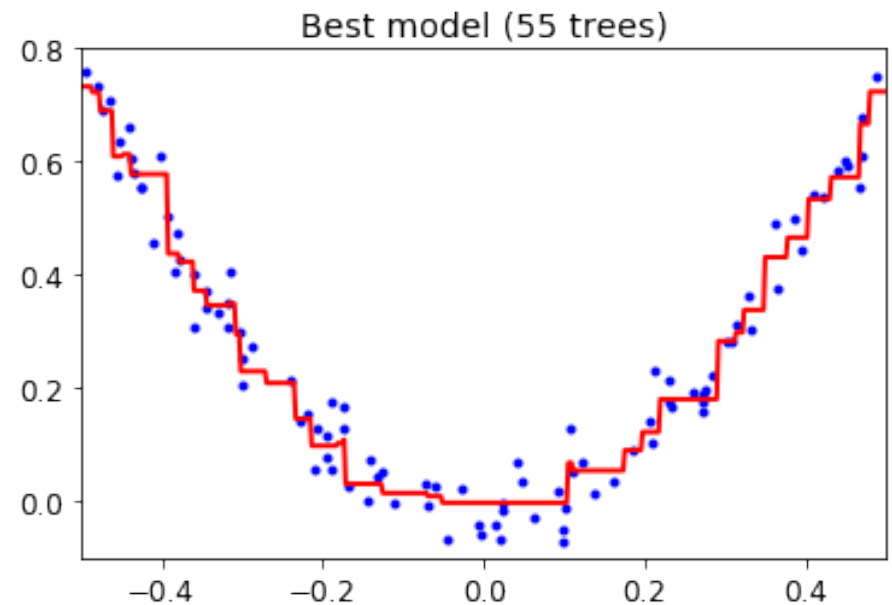
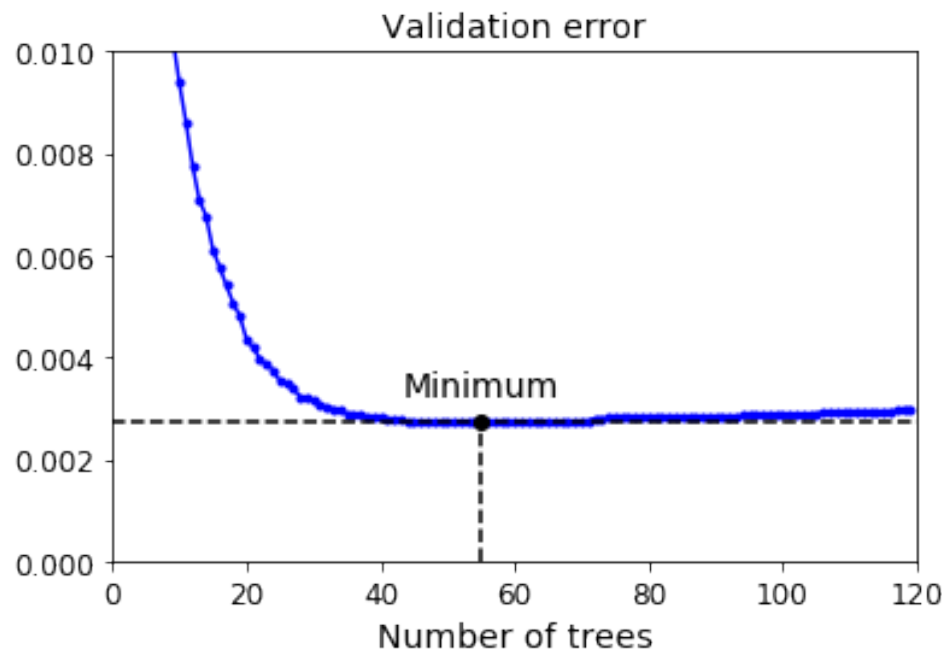
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- XGBoost stands for eXtreme Gradient Boosting
- It uses a cleverly chosen regularisation term to favour simple trees
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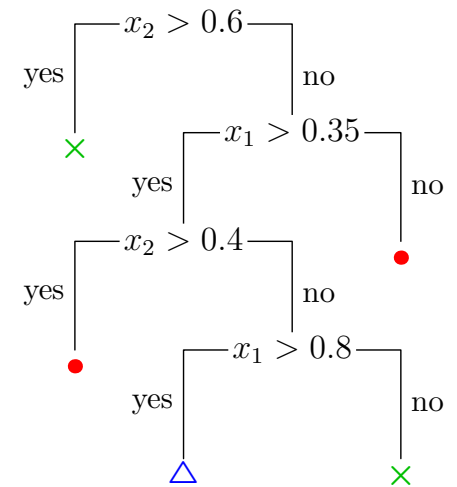
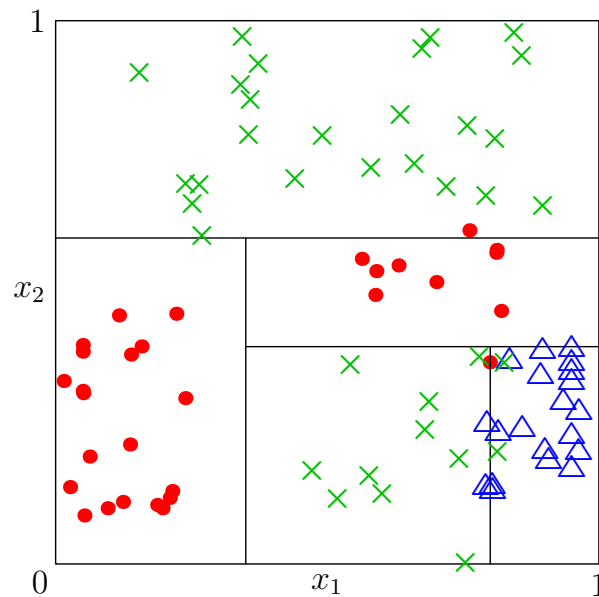
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Ensembling in Deep Learning

- For most machine learning ensembling different machines usually gives a reasonable improvement in performance
- The machines should have roughly the same performance
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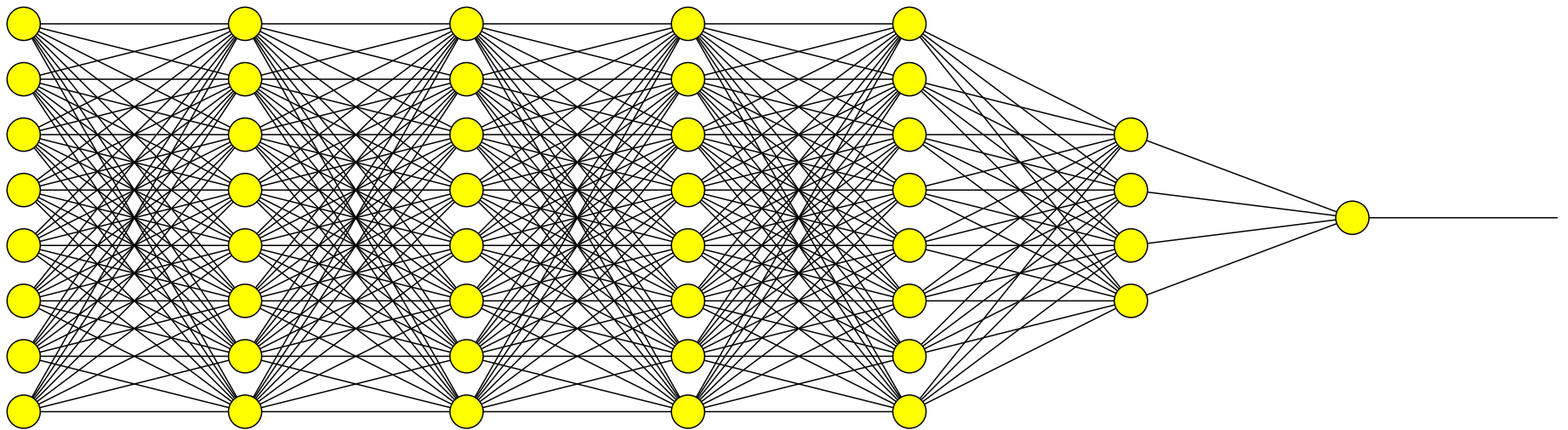
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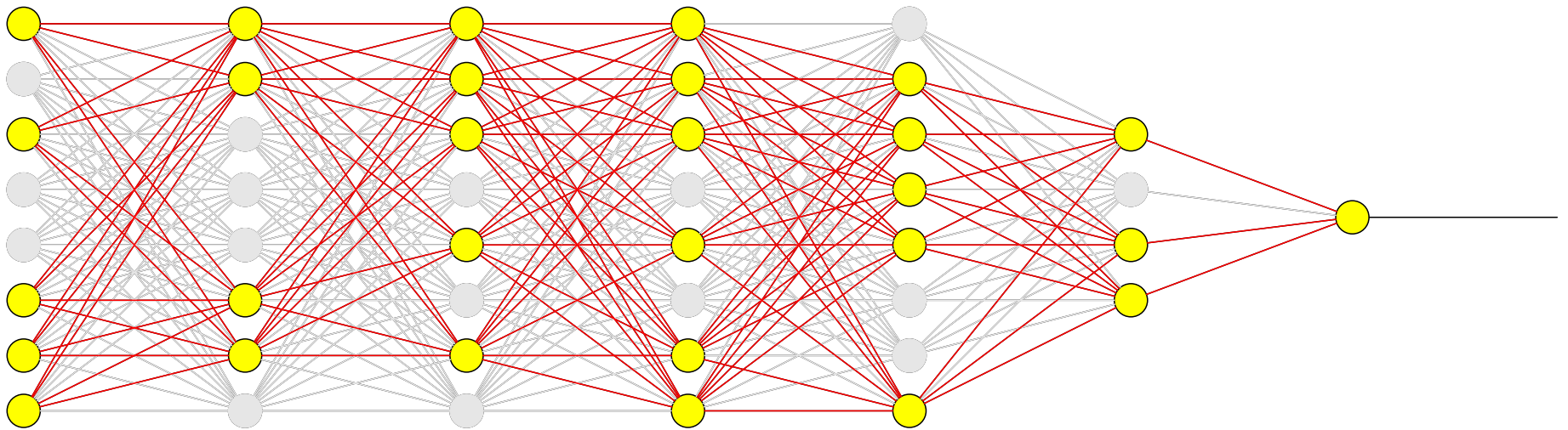
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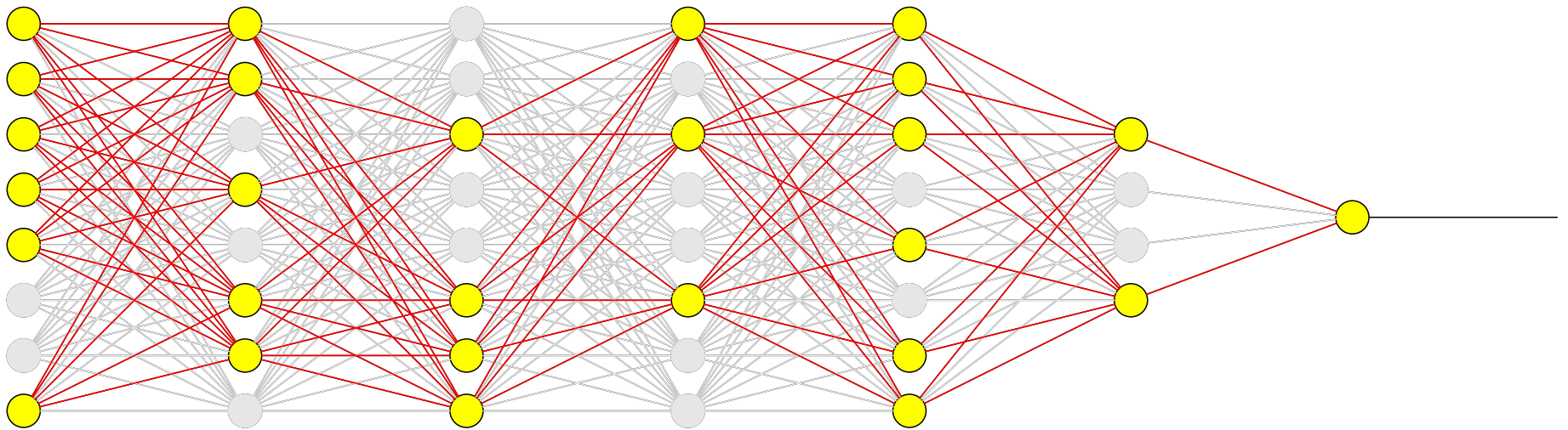
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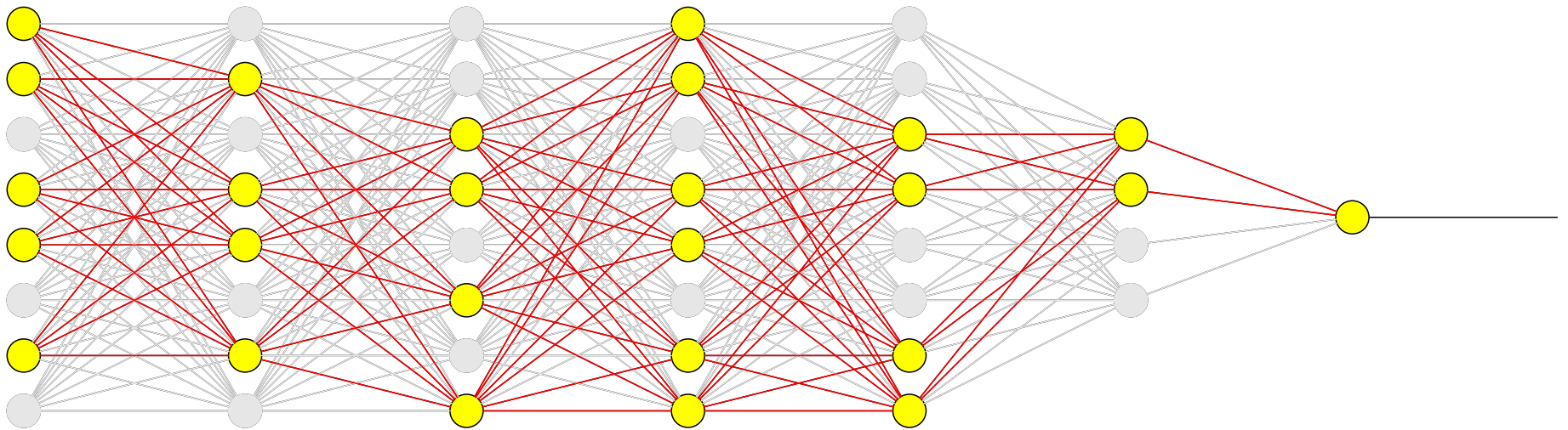
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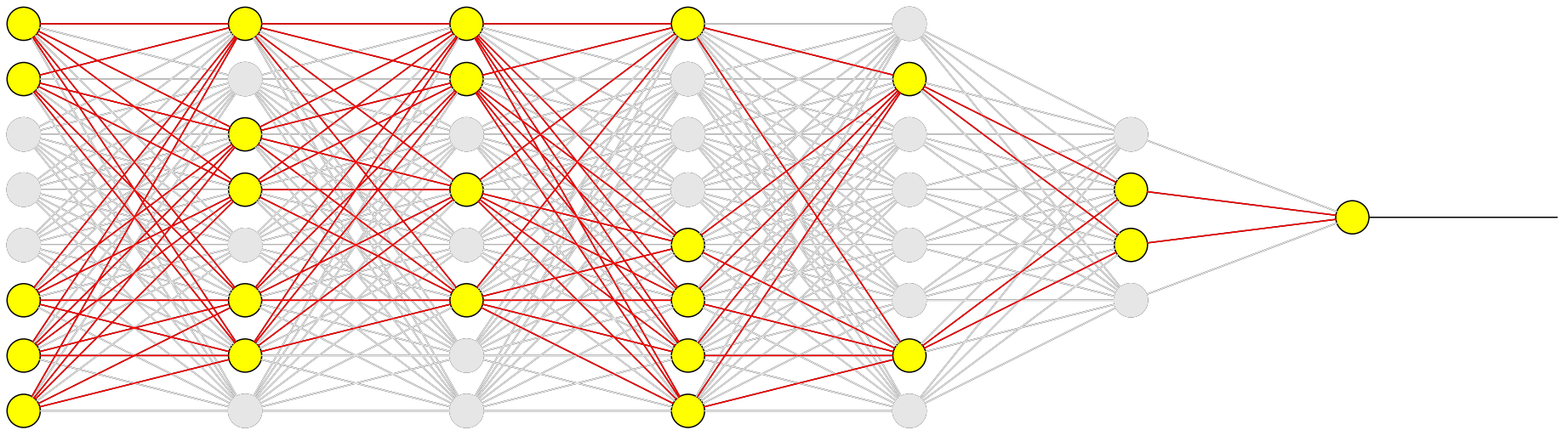
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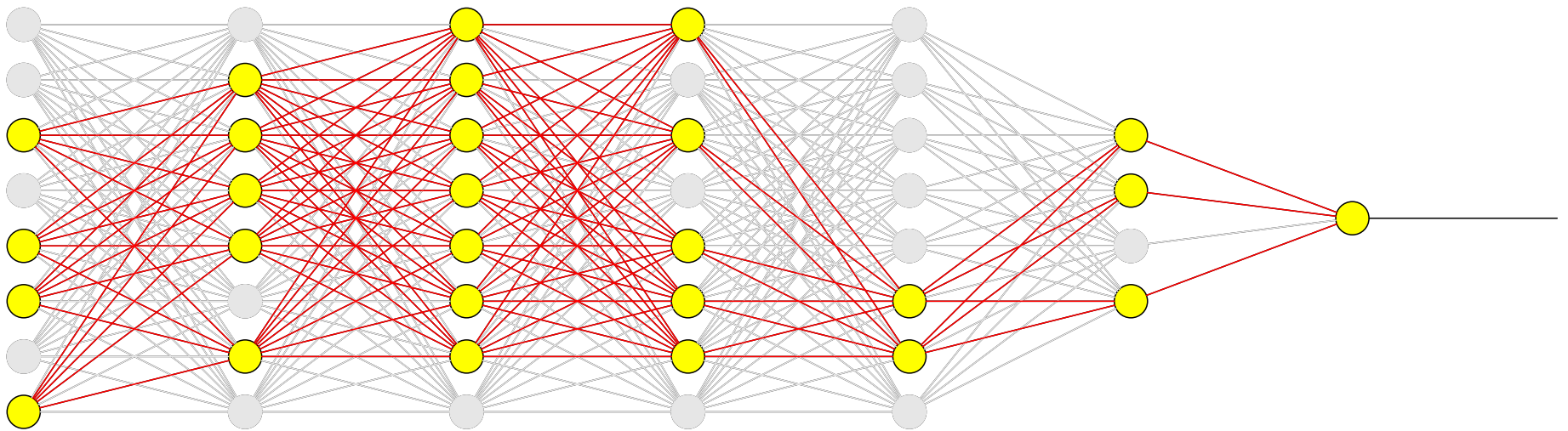
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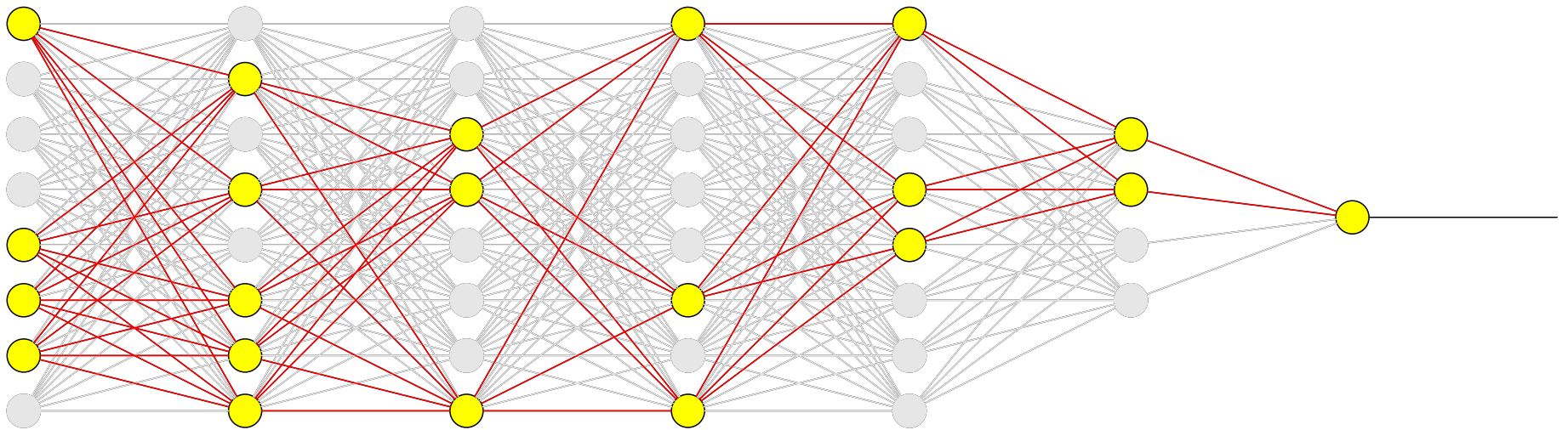
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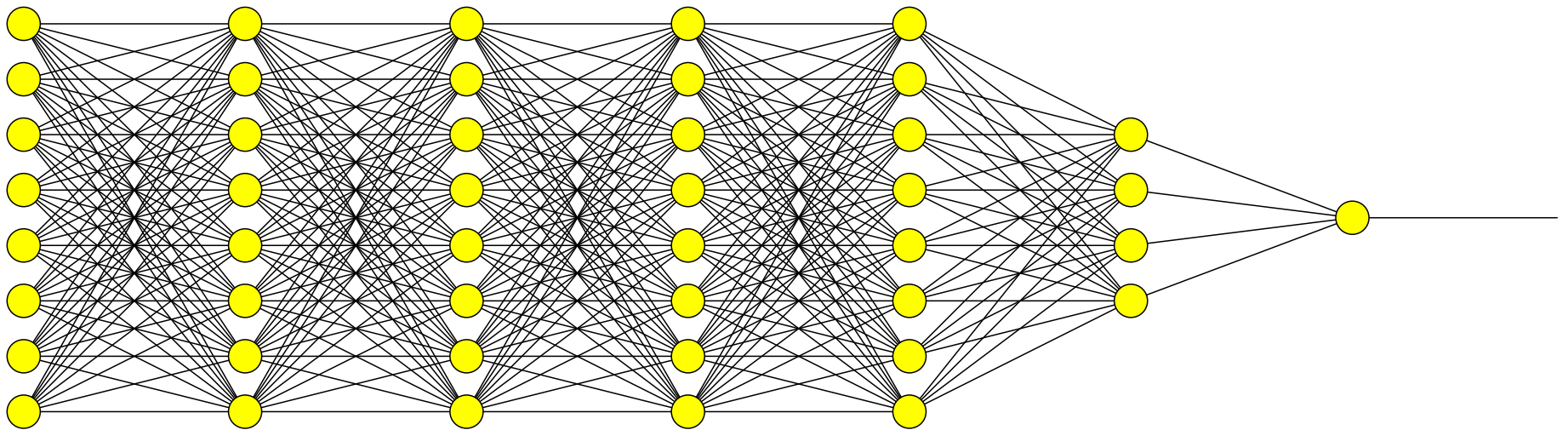
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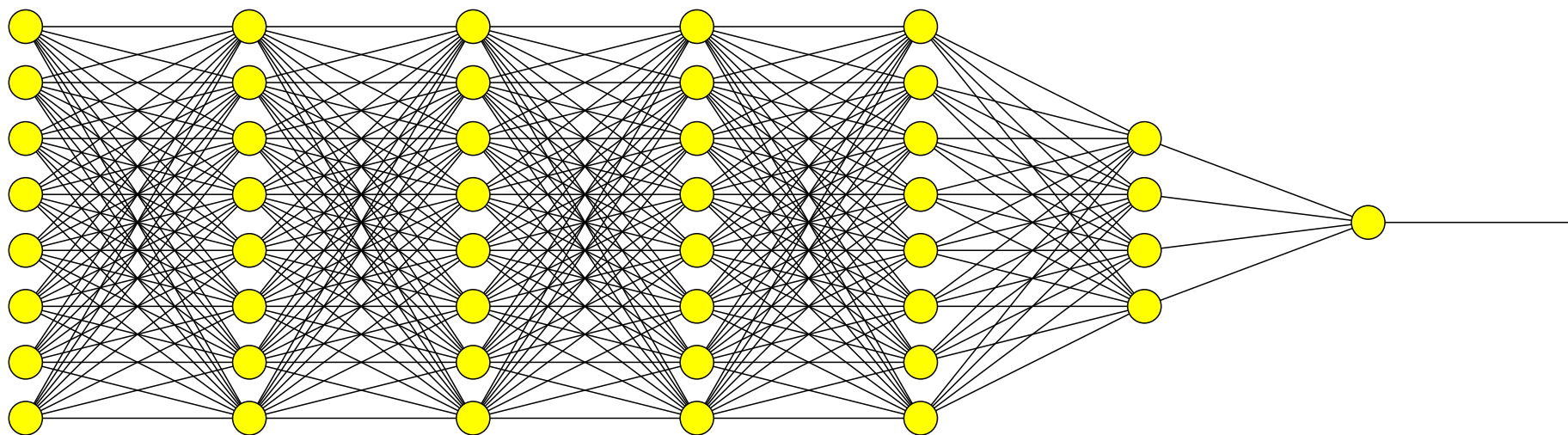
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- This can be seen as ensembling lots of much simpler machines

Conclusion

- Ensemble methods have proved themselves to be very powerful
- Tend to work best with very simple models (true of random forest and boosting)
- XGBoost or GBM are currently the best methods for tabular data (particular for large training sets)
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